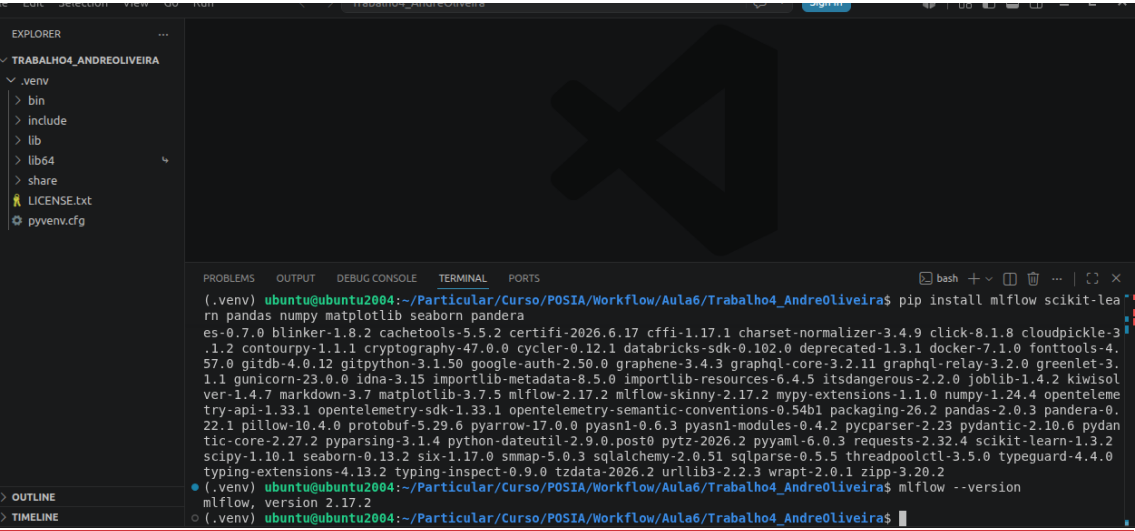


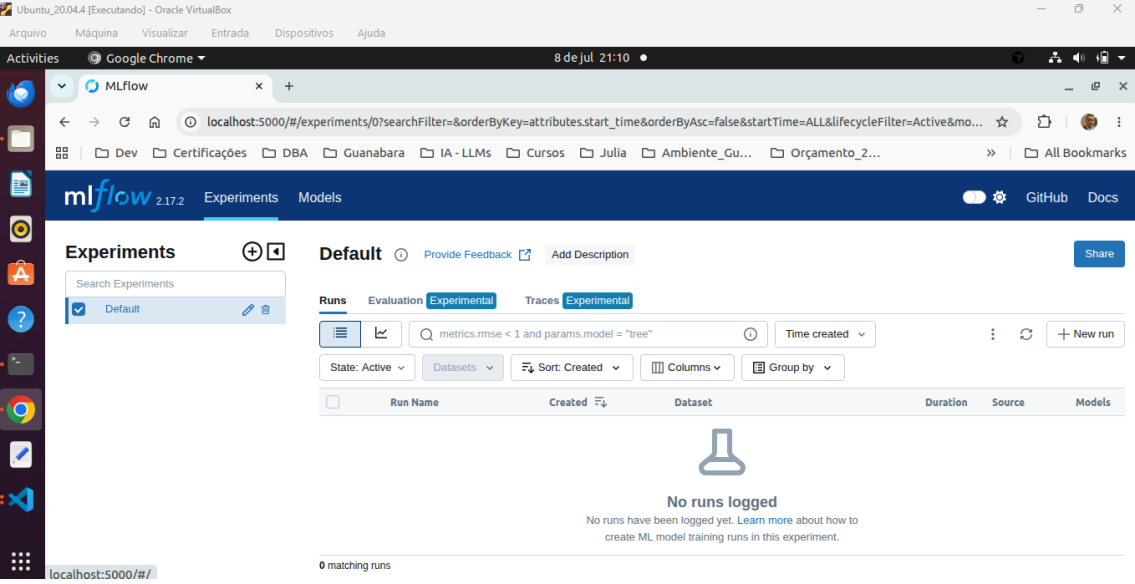
PucMinas
Curso: Engenharia de IA
Nome: Andre Cardoso de Oliveira

Matéria: Orquestração de Workflow
Trabalho: Atividade 4

Evidência 1



Evidência 2



Evidência 3

MLflow

2.17.2

Experiments

Models

Search Experiments

Default

wine-classification

wine-classification

Provide Feedback

Add Description

Share

Runs

Evaluation

Experimental

Traces

Experimental

metrics.rmse < 1 and params.model = "tree"

Time created

+ New run

State: Active

Datasets

Sort: Created

Columns

Group by

	Run Name	Created	Dataset	Duration	Source	Models
☐	GradientBoosting	4 minutes ago	-	2.2s	pipeline...	sklearn
☐	RandomForest	4 minutes ago	-	2.4s	pipeline...	sklearn
☐	LogisticRegression	4 minutes ago	-	6.9s	pipeline...	sklearn

3 matching runs

Evidência 4

MLflow

2.17.2

Experiments

Models

wine-classification

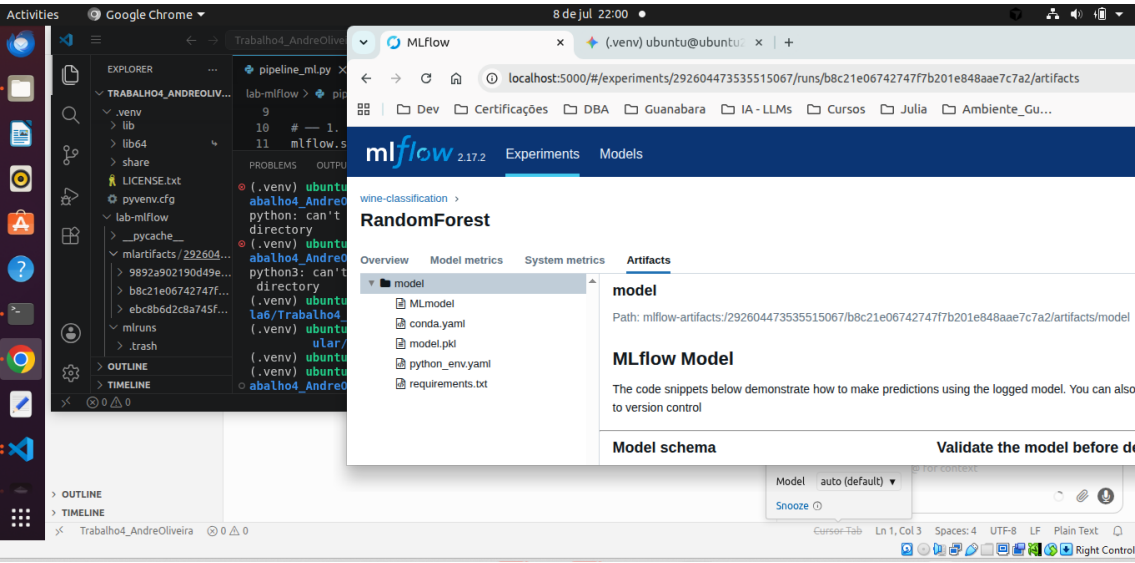
Comparing 3 Runs from 1 Experiment

Visualizations

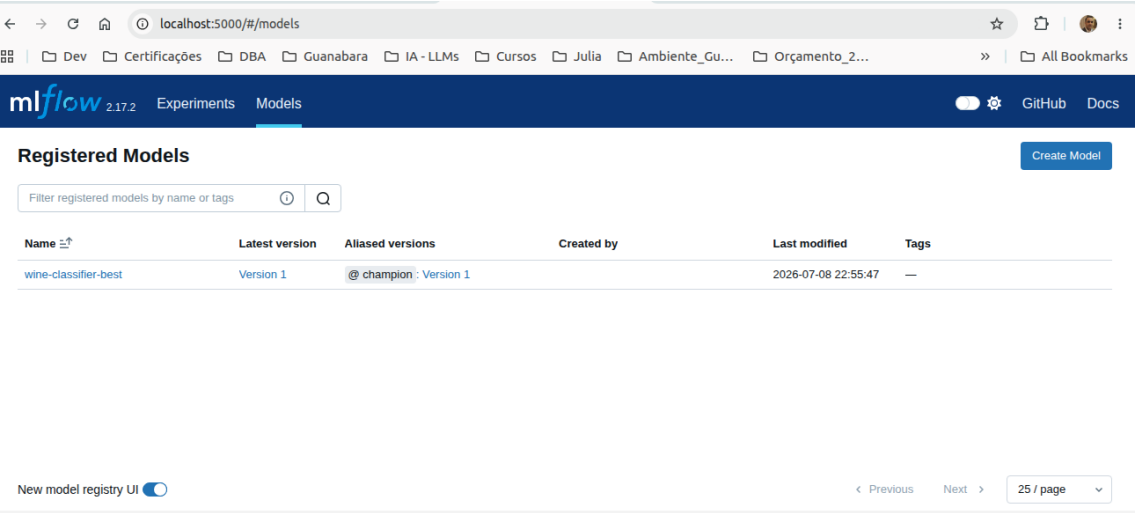
Run details

Run ID:	ebc8b6d2c8a745f09f20484bcb4573d	b8c21e06742747f7b201e848aae7c7a2	9892a902190d49e8ba8f0c43645f47f6
Run Name:	GradientBoosting	RandomForest	LogisticRegression
Start Time:	2026-07-08 21:36:24	2026-07-08 21:36:22	2026-07-08 21:36:15
End Time:	2026-07-08 21:36:26	2026-07-08 21:36:24	2026-07-08 21:36:22
Duration:	2.2s	2.4s	6.9s

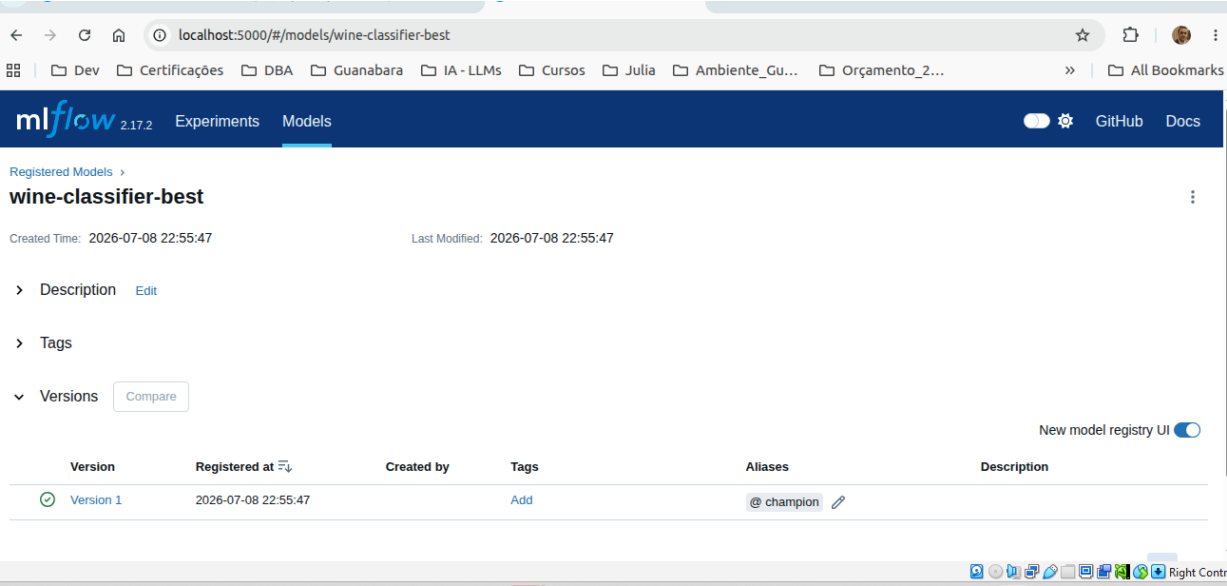
Evidência 5



Evidência 6



Evidência 7



Evidência 8

```
(.venv) ubuntu@ubuntu2004:~/Particular/Curso/P0SIA/Workflow/Aula6/Trabalho4_AndreOliveira/lab-mlflow$ ls
mlartifacts  mlruns  'pipeline_ml.py'  __pycache__  registrar_modelo.py  validar_dados.py
(.venv) ubuntu@ubuntu2004:~/Particular/Curso/P0SIA/Workflow/Aula6/Trabalho4_AndreOliveira/lab-mlflow$ python valida
r_dados.py
Dataset: 178 linhas, 4 colunas
count  alcohol  malic_acid  ash  alkalinity_of_ash
mean    13.00    2.34    2.37    19.49
std      0.81    1.12    0.27    3.34
min     11.03    0.74    1.36    10.60
25%     12.36    1.60    2.21    17.20
50%     13.05    1.87    2.36    19.50
75%     13.68    3.08    2.56    21.50
max     14.83    5.80    3.23    30.00

✓ Dados válidos – todos os checks passaram

--- Demonstração: dados com valor inválido ---
x Schema rejeitou os dados (esperado): Column 'alcohol' failed element-wise validator number 0: in_range(8.0, 16.0)
failure cases: 99.0
```

Evidencia 9

← → ↺ 🏠 🔍 localhost:5000/#/experiments/292604473535515067?searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTi... ☆ 🔄 🧑

🗄️ | 📁 Dev | 📁 Certificações | 📁 DBA | 📁 Guanabara | 📁 IA - LLMs | 📁 Cursos | 📁 Julia | 📁 Ambiente_Gu... | 📁 Orçamento_2... >> | 📁 All Bookmarks

mlflow 2.17.2 Experiments Models 🔌 ⚙️ GitHub Docs

Experiments (+) 🗑️

☐ Default

☒ wine-classification

wine-classification ⓘ Provide Feedback Add Description Share

Runs Evaluation Experimental Traces Experimental

Rename Delete Compare

	Run Name	Created	Dataset	Duration	Source
☐	RF-autolog	7 hours ago	dataset (528e06b1) Train	9.2s	autolog
☒	data-validation-failed	7 hours ago	-	37ms	validar
☒	data-validation	7 hours ago	-	155ms	validar
☐	data-validation-failed	16 hours ago	-	14ms	validar

8 matching runs

Evidencia 10

Google Chrome 9 de jul 09:14

MLflow x +

localhost:5000/#/compare-runs?runs=["8b638f632d58488fa34d373018ce53ac","b8c21e06742747f7b201e848aae7c7a2"]&experim...

🗄️ | 📁 Dev | 📁 Certificações | 📁 DBA | 📁 Guanabara | 📁 IA - LLMs | 📁 Cursos | 📁 Julia | 📁 Ambiente_Gu... | 📁 Orçamento_2... >> | 📁 All Bookmarks

Run Name:	RF-autolog	RandomForest
Start Time:	2026-07-09 08:23:50	2026-07-08 21:36:22
End Time:	2026-07-09 08:24:00	2026-07-08 21:36:24
Duration:	9.2s	2.4s

Parameters

☒ Show diff only

bootstrap	True
ccp_alpha	0.0

class_weight	None
criterion	gini
max_features	sqrt
max_leaf_nodes	None
max_samples	None

Evidencia 11

```
(.venv) ubuntu@ubuntu2004:~/Particular/Curso/POSIA/Workflow/Aula6/Trabalho4_AndreOliveira/lab-mlflow$ mlflow run
n . -P n_estimators=200 -P max_depth=8 --experiment-name wine-classification --env-manager=local
2026/07/09 19:23:59 WARNING mlflow.models.model: Model logged without a signature and input example. Please set
`input_example` parameter when logging the model to auto infer the model signature.
  accuracy=1.000 | f1=1.000 | run_id=9585d33829394dbdb4894f370ec1613c
2026/07/09 19:23:59 INFO mlflow.tracking._tracking_service.client: 🌟 View run RandomForest at: http://127.0.0.
1:5000/#/experiments/292604473535515067/runs/9585d33829394dbdb4894f370ec1613c.
2026/07/09 19:23:59 INFO mlflow.tracking._tracking_service.client: 🟢 View experiment at: http://127.0.0.1:5000.
/#/experiments/292604473535515067.

Treinando GradientBoosting...
2026/07/09 19:24:00 WARNING mlflow.models.model: Model logged without a signature and input example. Please set
`input_example` parameter when logging the model to auto infer the model signature.
  accuracy=0.944 | f1=0.944 | run_id=5ac8cfe834ab4eb7806654e97c1d528b
2026/07/09 19:24:01 INFO mlflow.tracking._tracking_service.client: 🌟 View run GradientBoosting at: http://127.
0.0.1:5000/#/experiments/292604473535515067/runs/5ac8cfe834ab4eb7806654e97c1d528b.
2026/07/09 19:24:01 INFO mlflow.tracking._tracking_service.client: 🟢 View experiment at: http://127.0.0.1:5000
/#/experiments/292604473535515067.

Treino concluído. Acesse http://localhost:5000 para ver os runs.
2026/07/09 19:24:01 INFO mlflow.projects: === Run (ID '6932964650724cb3946ff5b61c226d4a') succeeded ===
(.venv) ubuntu@ubuntu2004:~/Particular/Curso/POSIA/Workflow/Aula6/Trabalho4_AndreOliveira/lab-mlflow$
```

Evidencia 12

localhost:5000/#/experiments/292604473535515067/runs/9585d33829394dbdb4894f370ec1613c

mlflow 2.17.2 Experiments Models

wine-classification > RandomForest Register model

Overview Model metrics System metrics Artifacts

Parameters (3)

Search parameters

Parameter	Value
random_state	42
n_estimators	200
max_depth	8

Metrics (2)

Search metrics

Metric	Value
accuracy	1
f1_score	1